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PROJECT: DataVerse Week 1 Project

Hospital Operational Analysis Report

Objective: This analysis aims to identify key hospital performance metrics gotten from the dataset and possible insights gathered

No	Metrics	Definition	Why it matters?
1.	Average Length of Stay (LOS)	Mean duration between patient admission and discharge	Shorter LOS suggests effective diagnosis, treatment planning, professionalism and timely discharge. Longer LOS may reflect complications, high severity or inefficiencies
2.	Average ED Arrival of Provider Seen Time	Time between a patient's arrival in the Emergency Department and the first provider contact	Prompt response is critical for patient safety, especially for cardiac arrest, trauma, infections and CT head cases. Delays increase overcrowding and nosocomial infection risk.
3.	Re-admission Rate (30 days)	Percentage of patients who return within one month after discharge	Indicates quality of care and discharge effectiveness. High rates may suggest poor medication adherence, inadequate follow-up, unresolved conditions or premature discharge.
4.	Average Cost per Encounter	Mean financial cost associated with each patient visit.	Reflects resource allocation staffing, diagnosis, treatment and length of stay. Helps evaluate

			financial availability and identify inefficiencies.
5.	Distribution of Admission Types	Proportion of Emergency, Urgent and Elective admissions	Admission type affects predictive planning. Emergency requires immediate attention, urgent is prompt but non-life threatening, elective is planned and usually less severe
6.	Patient Severity by Admission Type	Distribution of severity levels across Emergency, Urgent and Elective admissions.	Elective cases have least severity due to proper planning. Emergency cases show highest severity, requiring rapid interventions, more resources and closer monitoring.
7.	Patient Outcome By Discharge Disposition	Relationship between admission type and patient outcome (Recovered, Expired, Skilled Nursing, Home)	Emergency admissions show higher expired and skilled nursing outcomes. Urgent cases have the highest return-home outcomes. Elective admissions show the least expired outcomes.
8.	Gender Distribution by Department	Proportion of male and female patients across departments	Females outnumber males overall. Males dominate Orthopedics, Surgery and Cardiology while females dominate Pediatrics and Emergency. Supports cost planning and departmental budgeting

9.	Average Cost by Departmental and Gender	Mean encounter cost broken down by department and patient gender	Internal Medicine and Orthopedics are more expensive due to complexity, diagnostics and longer LOS. Supports cost planning and departmental budgeting.
10.	Age Group by Outcome	Distribution of patient outcomes, across age groups.	Ages 30-40 represent the largest patient group. Helps plan staffing, preventive care and investigate prevalent ailments in this age range.

INSIGHTS SUMMARY

What's the Hospital doing well operationally?

Efficient Discharge Process: 74.7% of patients fully recovered and returned home

Prompt Provider Response in ED: Adequate staffing and functional triage improved patient safety

Well-Managed Elective Care: Low severity and re-admission rates indicate strong planning

Effective Emergency Workflow: Even with high emergency volume, outcomes remain reasonable.

Where Bottlenecks or Risks Likely Exist?

Re-admissions within 30 days: May indicate gaps in discharge education, follow-up, or medications adherence.

High-Severity Emergency Load: Sudden spikes could strain staff, equipment and bed capacity

ED Wait Time: Extended waits may reflect staffing shortages, limited imaging and lab resources or inefficient triage

Cost Variability Across Encounters: Suggests potential overuse of diagnosis or resource inefficiencies.

Results that should concern Leadership

High Re-admission Rates: Might signal possible quality gaps in treatment or discharge planning

Severe Emergency Cases with Poor Outcomes: Indicates patient safety risk and possible capacity limits

Long ED Wait Times: Impacts patient satisfaction and safety

High Cost, High-Frequency Procedures: Drives workload and spending. E.g. Procedure 81002

Procedure and Diagnosis Cost Drives

Procedures 81002: Highest encounter count and cost. Key workload and financial driver

Diagnosis N39 (Most common): High demand for routine management and chronic care planning

Diagnosis A08: Less frequent but highest average cost. Cost-intensive, not volume-driven

Clarifying Questions Before Action

1. Are repeat visits by the same patient counted as separate encounters?
2. Are clinicians experiencing workload pressure affecting discharge timing?
3. Are delays caused by shared resources (labs, imaging)?
4. How is discharge planning conducted and what follow-up exists to reduce re-admissions?
5. Are staffing or equipment shortages contributing to ED delays or high-severity care strain?

Implement post-discharge follow-up for high-risk patients to reduce re-admissions

Review cost standardization protocols especially for high-cost procedures like 81002

Consider predictive staffing models for peak ED periods

Thank you